

HiVideo: Hierarchical Browsing Interface for Educational Videos

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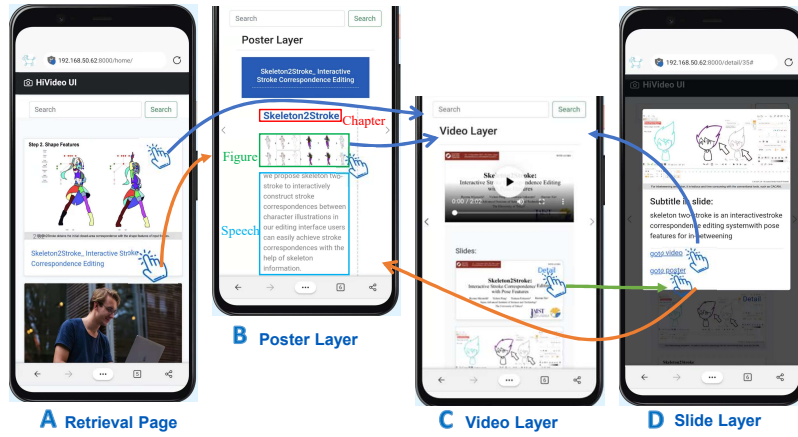


Figure 1: The proposed HiVideo interface with retrieval page (A), poster layer (B), video layer (C) and slide layer (D).

CCS CONCEPTS

• **Human-centered computing** → *Interaction design*.

KEYWORDS

slide-based video, user interface, information retrieval

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1 INTRODUCTION

Online academic conferences and massive open online courses (MOOCs) are becoming increasingly popular. However, it is difficult to properly retrieve and select the required video material. Current video retrieval systems [Kim et al. 2014; Yadav et al. 2016] may require more interactivity and summarization features to meet

users' browsing habits. For example, if a user wants to find a certain approach at an academic conference, they will most likely obtain multiple results with video thumbnail and title after long-time searching. These interfaces may not allow users to intuitively know the video contents.

We propose a video browsing interface, HiVideo, for academic presentations and educational videos, as shown in Fig. 1. When users retrieve the specified contents, the poster layer can help them explore the general video content rather than watching the whole video. To extract slides from a video, the proposed system leverages a frame detection approach, and a deep learning based document layout analysis approach [Shen et al. 2021] to extract visual information from the slides. In the user study, we verify that this proposed HiVideo interface can accommodate a wide range of usage scenarios for different users. In contrast to a previous video browsing interface [Weng et al. 2022], we adopt responsive web design so that users can achieve the better user experience even with mobile browsers.

2 SYSTEM OVERVIEW

System Framework. Our video dataset consists of 20 educational academic videos and the corresponding video content extracted from video pre-processing, which is a fundamental stage in cataloging and summarizing video material as shown in Fig. 2. We first compared the image hash of each frame in an educational

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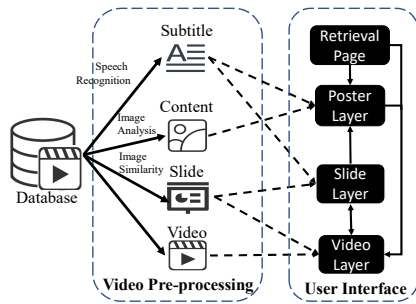


Figure 2: The system framework of the proposed HiVideo.



Figure 3: Results of the comparison study. The ordinate denotes the accuracy of correspond user interface.

video. If the difference between the two frames is greater than the threshold (2 in this work), the slide will be altered. We adopted a deep learning-based document analysis that includes a collection of layout data structures and APIs designed for document layout processing [Shen et al. 2021]. We conducted the extraction process to obtain titles, figures, and tables from the retrieved slides.

Preliminary Study. We conducted a preliminary investigation (15 college students around 20 years old, 12 males and 3 females) to explore why users are not satisfied with the existing interface and their search habits. We found that when users search for a certain topic, they prefer to use textual content or graphics in their slides. To solve these issues, we propose a hierarchical video browsing interface with poster, slide, and video layers.

User Interface. In our browsing interface, we created a poster layer that displays the video chapter on slides and the spoken material on the video in poster format. The poster layer can be interactively navigated to other two layers as shown in Fig. 1(B). To allow users to associate the video with the slide in a convenient way, we designed the video layer as illustrated in Fig. 1(C). The slides on the right will change in sync with the video as it plays. A ‘click’ operation on the slide allows users to fast-forward the video to the specific segment they want to browse. To solve the issue in the preliminary study, users can choose to read text instead of video, and we designed the slide layer. As shown in Fig. 1(D), users can view the details (textual matter of the spoken content of the slide) of the specific slide. In addition, users can navigate to other layers using ‘click’ operation.

Question	Score	Mean	SD
How easy for you to use the system without any instructions?		5.38	0.74
Do you think this UI can help you summarize the video?		5.63	0.74
Does this 3 layer help saving time when retrieve and skim videos?		6.75	0.46
Do you think the Poster Layer helps you to retrieve the video?		6.50	0.54
Do you think the Frame Layer helps you to retrieve the video?		6.13	0.35
Do you think the Video Layer helps you to retrieve the video?		5.88	0.84
What is your overall feeling for this system?		5.75	0.71

Figure 4: Results of the questionnaires in user study.

3 USER STUDY

We conducted the comparison study to compare the traditional retrieval interfaces to our interface. The compared interfaces are the video-only style and the Coursera style (video-only style with the narration underneath.). We divided 15 participants into three groups and asked them to retrieve 5 certain videos with the given requirements in 5 minutes. After the experiment, we calculated the accuracy value taking the average score of each interface. Fig. 3 shows that our interface achieved the highest accuracy results. Then, we conducted a subjective evaluation to validate the user experience by asking participants to complete three video retrieval tasks: visual search, question search, and summarization. For visual search task, we asked participants to locate specified contents in the video, such as “Find a slide where the author discusses the research framework.”. For the question search task, we asked participants to answer a video-related question, such as “What is the input in the partial-editing layer?”. For summarization task, the participants were asked to summarize the essential topics in the video. Finally, a questionnaire was conducted about the user experience as illustrated in Fig. 4. The results confirmed that HiVideo can achieve high retrieval accuracy and provide a good user experience.

4 CONCLUSION

In this work, we proposed HiVideo, a hierarchical browsing interface to retrieve and summarize instructional education videos. We adopted a three-layer user interface to satisfy users’ searching intents in a variety of usage circumstances. Additionally, we conducted a user survey to validate our interface. In the future, it is promising to improve the algorithm to adapt to multiple types of video so that it can increase the usability of the retrieval interface and meet diverse retrieval intentions. We plan to improve the summarization algorithm to adapt to long-duration videos.

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